FH6SmartShift (v1.0.0)

Virtual Dyno & Shift Telemetry System for Forza Horizon 6 — User & Technical Manual

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Build: v1.0.0
Monolith
Sampling: 60 Hz
UDP

Platform: SimHub | .NET Framework 4.8 | C# 7.3 | License: GPLv3

1. Overview & Core Architecture

FH6SmartShift is a real-time C# telemetry plugin for **SimHub** and a dedicated racing dashboard engineered for the **Forza Horizon 6** physics engine. Instead of relying on static redline percentages, the plugin constructs a live **401-band Virtual Dyno** (`50 RPM` resolution up to `20,000 RPM`) and calculates mathematical power curve intersections across Gears 1–10.

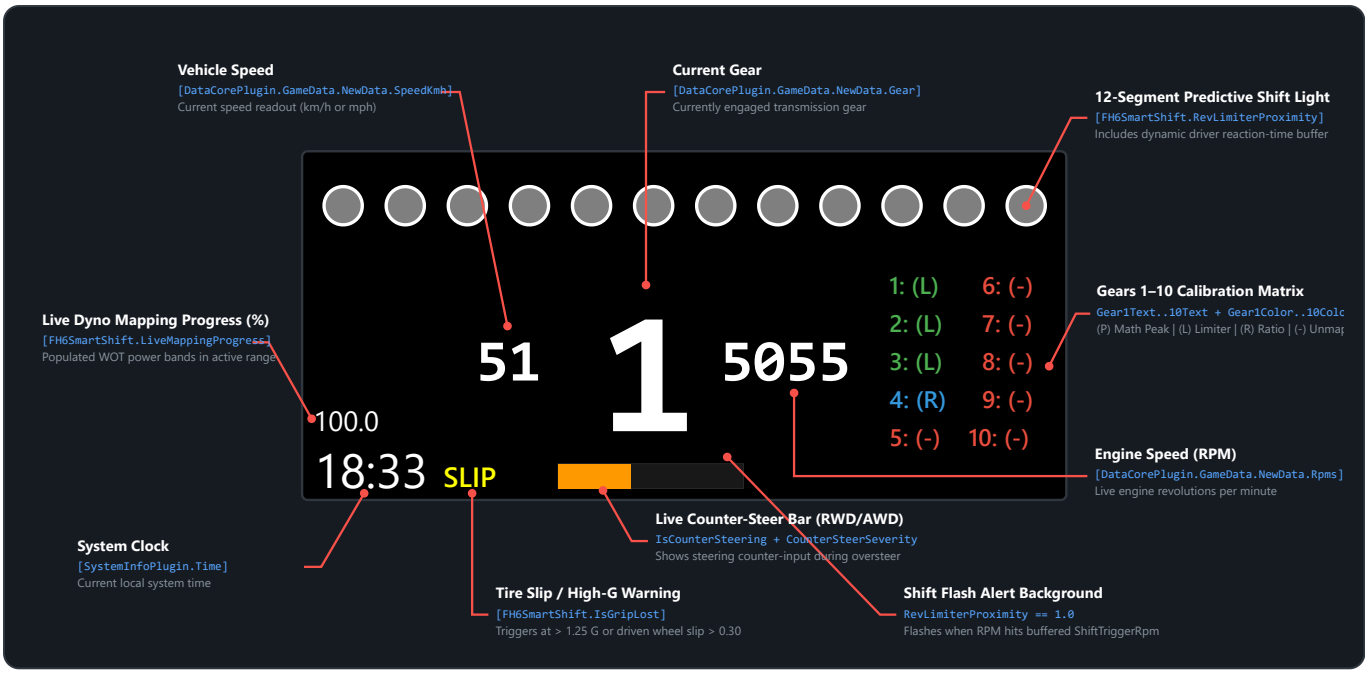
- **401-Band Virtual Dyno:** Samples engine power, torque, and turbo boost at wide-open throttle (`Accel >= 242` , ~95% throttle, `> 5000 W`) using an adaptive EMA filter.
- **Power Curve Intersection (`(P)`):** Compares current gear power (`pNow`) against next-gear landing power (`pNext` , including a `0.85` turbo boost-drop penalty if boost drops below 70%). Locks optimal shift when `pNext >= 1.015 * pNow` across 6 steps (25 RPM each) with `>= 85%` band density.
- **Autonomous Ratio Learning (`(R)`):** Locks gear ratios after 15 WOT samples (within a 25-sample buffer) when wheel slip is `< 0.10` and speed is `> 20 rad/s` . Automatically resets ratios if tuning changes are detected.
- **Ghost Limiter Detection (`(L)`):** Detects hidden power cuts before nominal `EngineMaxRpm` , purges contaminated dyno bands, and unlocks automatically if the engine later revs past the ceiling.
- **Multi-Layer Physics Filters:** Automatically pauses dyno learning during wheel lift (`<= 0.02`), puddles (`> 0.01`), damage (`> 0.05`), rear suspension spikes (`> 0.08 m`), surface rumble, clutch/handbrake use, or excessive wheelspin.

2. Quick Installation Guide

1. **Close SimHub completely** (running SimHub locks `.dll` files).
2. Copy `FH6SmartShift.dll` into your main SimHub root directory (default: `C:\Program Files (x86)\SimHub\`).
3. Start SimHub and enable the **FH6SmartShift** plugin when prompted.
4. Import the `Dashboard/` folder into SimHub Dash Studio (double-click the `.simhubdash` file or copy to `C:\Program Files (x86)\SimHub\DashTemplates\`).
5. In Forza Horizon 6 (**HUD and Gameplay** → **Data Out**), enable UDP telemetry and match the IP and Port configured in SimHub.

Database & Backups: Stored in `C:\Program Files (x86)\SimHub\PluginsData\Common\FH6SmartShift_fh6SmartShiftingData.json` (keeps 3 rolling daily backups `.bak_yyyyMMdd`).

3. Dashboard Visual Reference & Indicator Guide


Vehicle Speed
[DataCorePlugin.GameData.NewData.SpeedKmh]
Current speed readout (km/h or mph)
Current Gear
[DataCorePlugin.GameData.NewData.Gear]
Currently engaged transmission gear
12-Segment Predictive Shift Light
[FH6SmartShift.RevLimiterProximity]
Includes dynamic driver reaction-time buffer
Live Dyno Mapping Progress (%)
[FH6SmartShift.LiveMappingProgress]
Populated WOT power bands in active range
Gears 1–10 Calibration Matrix
Gear1Text..10Text + Gear1Color..10Colc
(P) Math Peak | (L) Limiter | (R) Ratio | (-) Unmap
51
1
5055
1: (L)
2: (L)
3: (L)
4: (R)
5: (-)
6: (-)
7: (-)
8: (-)
9: (-)
10: (-)
100.0
18:33
SLIP
Engine Speed (RPM)
[DataCorePlugin.GameData.NewData.Rpms]
Live engine revolutions per minute
Live Counter-Steer Bar (RWD/AWD)
IsCounterSteering + CounterSteerSeverity
Shows steering counter-input during oversteer
System Clock
[SystemInfoPlugin.Time]
Current local system time
Tire Slip / High-G Warning
[FH6SmartShift.IsGripLost]
Triggers at > 1.25 G or driven wheel slip > 0.30
Shift Flash Alert Background
RevLimiterProximity == 1.0
Flashes when RPM hits buffered ShiftTriggerRpm

4. Gears 1–10 Calibration Matrix States

State	Color / Hex	Actual Condition in C# Code
`N: (P)`	Green `#4CAF50`	**Peak Optimal (`IsConfirmed && IsMathOptimal`):** Confirmed power curve intersection with gear `N+1` (`pNext >= 1.015 * pNow`) and target band density `>= 85%` .
`N: (L)`	Green `#4CAF50`	**Limiter Confirmed (`IsConfirmed && TriggerIsGhost`):** (1) `GhostLimiterRpm > 0` detected near ceiling, or (2) highest known gear revved at WOT to `actualLimiter - 100 RPM` .
`N: (L?)`	Orange `#F39C12`	**Edge / Unconfirmed Limit (`TriggerIsEdge`):** Curves did not intersect before cutoff on a gear with `N+1` available. Provisional trigger: `actualLimiter - 250 RPM` .
`N: (P?)`	Yellow `#F1C40F`	**Unconfirmed Trigger:** Gear has a saved shift point (`ShiftTriggerRpm > 0`), but lost confirmation (e.g., after breaking through a ghost limiter).
`N: (R)`	Blue `#3498DB`	**Ratio Locked (`RpmPerRadS > 0`):** Gear ratio learned at WOT (`>= 15` samples); waiting for gear `N+1` ratio or top-gear WOT pull to limiter.
`N: (-)`	Red `#E74C3C`	**Unmapped / Paused:** Gear ratio not learned yet, memory reset, or game is paused / in garage menu (`IsRaceOn == 0`).

5. SimHub Telemetry Properties & Actions Reference

Property (`[FH6SmartShift.*]`)	Type	Description & Exact Behavior
`ShiftTriggerRpm` / `MaxUsableRpm`	`double`	Shift Light trigger reduced by dynamic driver reaction-time buffer (up to `1500 RPM` in G1–2, `600 RPM` in G3+).
`OptimalShiftRpm`	`double`	Unbuffered math shift point for `(P)` gears, `actualLimiter - 250` for `(L?)` , or `ShiftTriggerRpm - 50` for `(L)` .
`DownshiftTriggerRpm`	`double`	Safe downshift RPM raised by braking (up to `+600 RPM`) and front dive (`+250 RPM`), capped by limiter ceiling.
`RevLimiterProximity`	`double`	`0.0–1.0` LED progress. Rises from 75% RPM range; jumps to `1.0` when buffered `ShiftTriggerRpm` is reached at WOT.
`LiveMappingProgress`	`double`	Percentage (`0.0–100.0`) of populated dyno bands from `max(1000, 0.5 * RpmAtMaxPower)` to `actualLimiter - 250` .
`MapCoverageBar` / `EngineMapStatus`	`string`	20-char ASCII bar up to `actualLimiter` / Status string (`"OK"` at `>= 95%` , else `"LEARN"`).
`RpmAtMaxPower` / `RpmAtMaxTorque`	`double`	RPM at which peak smoothed power and peak smoothed torque were recorded.
`IsLearningComplete` / `IsTopGear`	`bool`	`true` when current gear is calibrated / `true` when confirmed top gear (requires `currentGear >= 5`).
`IsDownshiftSafe` / `IsDownshiftSuggested`	`bool`	Safe downshift flag / Downshift suggested under braking or load hill-climb assistant (`_climbDownshiftActive`).

Property (`[FH6SmartShift.*]`)	Type	Description & Exact Behavior
`IsTractionGood` / `IsGripLost`	`bool`	Dyno traction gate flag / Dashboard `SLIP` indicator (`> 1.25 G` or driven wheel slip `> 0.30`).
`PowerOversteerWarning` / `OversteerSeverity`	`bool/dbl`	Rear slip warning under WOT / `0.0–1.0` rear oversteer scale for RWD/AWD above `20 km/h`.
`IsCounterSteering` / `CounterSteerSeverity`	`bool/dbl`	Active in RWD/AWD when counter-steering against yaw rotation / Smoothed `0.0–1.0` steering counter-input.
`BodySlipAngle` / `AeroBottomingWarning`	`dbl/bool`	Chassis slip angle in degrees / Damper bottoming alert (`>= 0.98` travel above `200 km/h`).
`Gear1Text..10Text` / `Gear1Color..10Color`	`string`	Status labels and Hex color codes for Gears 1–10.

SimHub Actions: `FH6SmartShift.ResetPowerMemory` (resets current car dyno & gears) • `FH6SmartShift.ResetGearRatios` (resets only ratios/shift points after tuning) • `FH6SmartShift.ResetLimitGears` (resets only `(L)` and `(L?)` gears) • `FH6SmartShift.ResetAllCarsMemory` (full database wipe).

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⚠️ **Note:** `IsTopGear` requires `Gear >= 5` ; Counter-Steer indicators are disabled for FWD vehicles.

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